How Can a Desk-Based Product System Support a Return-to-Reading Habit Loop and Motivation to Sustain Academic Reading Among Students Aged 18–24?

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ABSTRACT

Academic reading increasingly competes with digital environments that offer faster stimulation, novelty and immediate reward. Within Human–Computer Interaction (HCI) and interaction design, digital distraction is often addressed through blocking tools, productivity tracking or gamified focus systems. However, these approaches do not fully address why students disengage from low-stimulation academic reading, whether engagement can be sustained through intrinsic motivation rather than forceful control, or how users recover reading context after attention shifts. This project therefore asks: how can a hybrid physical–digital reading experience help students aged 18–24 sustain engagement with academic reading and recover from digital distraction?

The research adopts a practice-led Research through Design methodology, combining theory with iterative making to examine how sensory and interactive qualities can shape the reading experience. Literature review establishes the theoretical foundation across digital distraction, task re-entry, motivation, flow and affective haptics. This is extended through precedent analysis, material experimentation, interaction studies, prototyping and critical reflection.

The design develops across three connected layers. The first explores a speculative dream study environment that also functions as a virtual exhibition space. The second translates satisfying physical behaviours, such as squishing, resistance, compression and recovery, into digital interaction to explore a more engaging and less stressful reading experience. The third introduces attention-guiding and task-re-entry mechanisms, including friction, calming interaction, history cues and assistance cues.

The research argues that academic reading support should move beyond simply restricting distraction. It proposes sustained engagement as the primary intervention, with task re-entry retained as a recovery mechanism. This is expressed through a redesigned habit loop: distraction → friction → calming interaction → contextual re-entry cues → return to the more engaging reaWing experience. The final outcome combines a speculative Reading Machine, virtual 3D exhibition environment and research publication. As formal large-scale user testing was outside the project scope, the outcome is positioned as an exploratory design proposition rather than a validated intervention.

1 INTRODUCTION

This project investigates how interaction design can support students aged 18–24 in returning from low-effort digital distraction to high-opportunity-cost study task – Academic reading.In this research, high-opportunity-cost tasks refer to activity where choosing one action means giving up the immediate rewards of another, often more stimulating alternative. In academic reading contexts, rewards are delayed, and where sustained effort is needed before meaning, understanding or achievement becomes clear thus easily interrupted by low-effort digital media, which can offer quicker stimulation, novelty and relief. This leads to a cycle that students are increasingly bored to high-oppunity-cost task(Tam and Inzlicht, 2024).

Because everyday study involves interruptions such as notifications, social surroundings, domestic demands, mental fatigue or necessary breaks, distraction cannot always be fully anticipated or avoided (Schneegass et al. 2022). This exegesis therefore rather than assuming distraction is an attention span issue, it reframes it as a habit loop problem from task re-entry problem to loss of motivation then at the end distracted by other things with lower opportunity cost. Task re-entry refers to returning to an interrupted task, a process made difficult because interruption displaces the mental context needed to continue (Altmann and Trafton 2007). In academic study, the student may need to recover the article, paragraph, argument, note or next sentence before work can resume. Re-entry is therefore not simply “paying attention again”; it is reconstructing the pathway back into the task. Therefore this researcher driven, research by practise project can be understand as researches the theory of redirect an existing distraction routine into a repeatable return-to-task habit loop and practise it through an AI-assisted, desk-based haptic product system that has combination of functional digital prototypes and speculative haptic devices.

1.1 Disciplinary positioning

This project sits across interaction design, Human–

Computer Interaction, behaviour-change design, affective haptics, educational technology and applied AI interaction. HCI provides a broader framework for studying how people interact with technologies, objects and digital environments. Behaviour-change theories explain how distraction can become a repeated routine and support my direction of using another designed behaviour to substitute that routine. This substitute behaviour is likely to be supported by affective haptics, as it explores how touch can operate as an emotional and sensory channel (Eid and Al Osman, 2016), which informs the calming cue after distraction. Educational technology research explains how motivation can help users re-enter flow, making the return to study feel more achievable, intrinsic and sustained without relying on forceful control.

There also have been proved precedent of people already investigate into this field, and these have also been analysed to help identify the strengths and limitations of existing products, so that this project can build on them critically. Forest is an app where users set a focus timer for themselves and are rewarded with tree growth once the session is completed. Although the gamified idea remains interesting, it creates a paradox because the app itself may become another distraction when users over-focus on the tree-growing progress (You, 2024). Tactile breathing devices such as Moonbird and Core show a different direction, where guided breathing can help users slow their breathing rate and reduce stress, often through a target rhythm of around six breaths per minute. However, they also show limitations because not all users can easily follow the same breathing guide (Honinx et al., 2025). This suggests an opportunity for AI-assisted customisation. In addition, these devices can support calming, but they do not guide attention back to the original study task. The gaps in these projects shows that a single-form intervention may be insufficient for this problem, which leads me to develop a hybrid system through Buchanan's view of design as the organisation of products, actions and systems. In this project, the physical haptic artefact supports calming, while the on-screen system guides attention back toward task re-entry.

The research may directly benefit university students who struggle to return to study after distraction, while also offering useful insights for HCI researchers, interaction designers, education technology designers and wellbeing-oriented designers interested in less screen-dependent forms of AI-supported study assistance.

1.2 Article and Theory Overview

As a research by practise project the project has been developed in a way that theories research has been found, then respond it with practise, therefore you would see research and experiments and design intersect with each other. In section 2, it covers the fundamental theories of creating a habit loop ….. (finish later when everythings has been written)

1.3 Experiments and Final Outcome

After the theories has been analysed, I run many experiments and refinement to develop my design, in section (finish later when everythings has been written)

2 THEORIES OF HABIT LOOP

The theoretical foundation of this project is habit formation, which informs the exploration of a repeated behavioural loop that supports both task re-entry and sustained engagement. Habits develop through repeated behaviours associated with stable contexts, gradually requiring less conscious deliberation (Pinder et al., 2018; Gardner, Lally and Wardle, 2012). In the context of academic reading, an unwanted behavioural pattern may emerge when students encounter difficulty or boredom, turn towards low-effort digital distractions, and receive immediate stimulation or relief.

Rather than relying on distraction-blocking or forced attention, this project proposes an alternative habit loop that makes returning to reading more accessible and continued engagement more desirable. The proposed sequence consists of four stages: friction, calming interaction, task re-entry and sustained engagement.

The first two stages explore frictional feedback and calming interaction as speculative mechanisms for interrupting distraction and supporting an emotional transition towards reading. However, these are not the primary research focus. Instead, the project concentrates on the final two stages: task re-entry, which involves recovering the mental context necessary to resume reading, and sustained engagement, which investigates how interactive reading experiences, achievable goals and motivational feedback might encourage users to continue.

These two stages are interconnected. Successfully returning to a task does not necessarily prevent subsequent disengagement, while an engaging reading experience remains insufficient if users cannot recover their previous context after interruption. Therefore, the proposed habit loop aims to connect recovery and continued engagement within a repeatable reading experience.

2.1 Habit Formation

Furthermore, habit-formation research suggests that simple actions repeated in stable contexts are more likely to become automatic over time (Gardner, Lally and Wardle, 2012). For this reason, the physical artefact should be accessible and based on an existing desk object, such as a mouse, phone case or water bottle, because users may already have familiar habits around these forms. And through digital visual cues, it also

should be easy repeating action low-effort tactile actions, such as touching, pressing, holding or squeezing, are some of the important actions because they are easier to repeat and can remain peripheral. This aligns with behaviour-change research suggesting that habits, behavioural skills and access can be stronger targets for behavioural change (Albarracín, Fayaz-Farkhad and Granados Samayoa, 2024).

Habit formation therefore informs not only the design of accessible, repeatable actions, but also the broader intention to establish a consistent return-to-reading routine. By connecting contextual recovery with manageable reading progression, the proposed system explores whether repeated interactions could gradually make task re-entry and continued engagement more familiar and less effortful.

2.3 Importance of Attention

Haptic interaction is further informed by Bakker and Niemantsverdriet's (2016) interaction-attention continuum, which distinguishes three types of interaction: focused, peripheral and implicit. Focused interaction requires deliberate, conscious attention, whereas peripheral interaction occurs at the edge of awareness while attention remains primarily directed towards another activity. Implicit interaction occurs when a system interprets users' everyday behaviours as input without requiring intentional interaction. This research primarily investigates peripheral interaction, as academic reading already demands substantial visual and cognitive attention. For example, checking a smartphone requires focused attention and interrupts reading, whereas a simple, familiar tactile action may operate more peripherally without significantly competing with the primary task.

Nevertheless, implicit interaction presents opportunities for adaptive and personalised feedback. Macdonald et al. (2021) identified substantial differences in individuals' preferences for calming haptic sensations, while Blanc et al. (2024) observed variations in participants' spontaneous squeezing frequencies and recommended adapting feedback to users' natural movements. These findings suggest that personalised interactions may better accommodate individual habits, preferences and sensory needs. Building on this principle, AI-assisted systems could potentially recognise interaction patterns and automatically adjust feedback to individual users. However, the effectiveness of such adaptation for academic reading remains unverified. Due to technological constraints, implicit personalisation will not be the primary focus of this project, although some conceptual explorations will consider its potential. Instead, the research prioritises low-demand peripheral interaction, with implicit adaptation identified as an opportunity for future development.

Within the proposed habit loop, attention demands may vary between stages. Friction and calming interactions are intended to operate with limited attentional demands, while contextual recovery and interactive reading activities may require brief periods of focused interaction. The design challenge is therefore not to eliminate focused attention, but to ensure that supporting interactions do not unnecessarily compete with academic reading, which remains the primary activity (Bakker and Niemantsverdriet, 2016).

2.4 Stage 1 of Habit Loop - Friction

As the first stage after attention has drifted, frictional feedback makes the moment noticeable. Frictional feedback can interrupt habitual behaviour when it is meaningful and acceptable rather than purely annoying (Laschke, Diefenbach and Hassenzahl, 2015). In this project, friction functions as an attention-guiding action of the artefact. It interrupts the automatic movement toward distraction, but should not be so strong that it becomes another source of pressure. This could be achieved through specific haptic movements, such as gentle vibration, slow left-to-right movement across the device, or subtle rotational movement. The aim is to borrow the user's main visual attention only briefly, redirecting attention toward the haptic artefact before guiding the user back into the study task instead of allowing them to continue their usual distraction routine.

2.5 Stage 2 of Habit Loop - Calming technology

After friction makes drift noticeable, the system should shift into calmer support. Research on calming haptic preferences suggests that users often prefer soft textures, smoothness, animal-like touch and fabric-like sensations (Macdonald et al., 2021). For a desk-based artefact, these qualities need to be translated into practical materials rather than copied literally. Fur, velvet or microfiber may feel soft, but they can collect dust, absorb moisture and become difficult to maintain. A smooth elastomeric surface, such as silicone or TPU, better supports repeated handling while remaining durable and easy to clean. In terms of calming haptic movement, it is found that expansion and contraction were preferred over vibration in reduce respiratory rate. (Honinx et al., 2025). This supports the decision to prioritise breathing-like expansion and contraction as a key calming movement in the prototype, rather than relying on sharp notification-like vibration.

2.6 Stage 3 of Habit Loop - On Screen Task re entry cues

Task re-entry represents the first primary research focus within the proposed habit loop. Although frictional and calming interactions may support the transition away from distraction, they cannot independently restore the cognitive context required to resume academic reading. Therefore the screen is introduced to guide the attention back after friction has been made. The screen-based cue

layer then addresses the cognitive side of task re-entry after the user has become calmer. The difficulties of task re entry is that cognitive recovery following an interruption is gradual rather than immediate. When attention shifts away from a complex task, the mental context supporting performance becomes displaced, requiring individuals to progressively retrieve previous goals, information and intended actions. This process, known as incremental retrieval, contributes to resumption lag, during which performance is initially slower before returning towards normal (Altmann and Trafton, 2007). For example, when readers return to a text after being distracted, they cannot always resume immediately. Instead, they must gradually reconstruct their previous reading context: which book was I reading, which page was I on, which paragraph had I reached, and what was I trying to understand? Each recovered piece of information helps retrieve the next, allowing readers to rebuild their train of thought. If part of this context is missing, they may struggle to identify where to continue or understand how the information connects.

Therefore task-resumption cue research suggests that cues are more useful when they help users actively retrieve previous learning rather than simply display information (Schneegass et al., 2022). This cueing layer should provide both history cues and assistance cues. A history cue helps the user recover the previous task context, such as an earlier highlight, note, annotation or section of text where attention broke down. An assistance cue is helpful after the memory has been retrieved by points toward the next action after interruption, reducing the effort required to decide how to restart (Chen et al., 2025). The design implication is that re-entry support should answer two practical questions for the user: where was I, and what should I do next?

2.7 Stage 4 of Habit Loop - Enter flow state and maintain motivation

Sustained engagement represents the second primary research focus and the intended continuation of the proposed habit loop. Although contextual cues may help readers return to an interrupted task, successful re-entry does not necessarily ensure continued engagement. Students may recover their reading position yet disengage again if the material remains difficult, monotonous or insufficiently rewarding. Therefore, supporting task re-entry alone may address the immediate consequences of distraction without adequately addressing repeated disengagement. Research on flow suggests that clear goals, appropriately matched challenges and meaningful feedback can support deeper engagement with learning activities (Hui et al., 2025). Similarly, research on student motivation highlights the importance of intrinsic motivation and personally valued goals in supporting persistence and positive learning outcomes (Howard et al., 2021). These findings inform the exploration of interactive reading experiences that make academic content more manageable, provide visible progression and introduce meaningful opportunities for interaction.

Rather than treating continued reading as a separate outcome, this project positions sustained engagement as an essential component of the habit loop. Task re-entry enables readers to resume, while engagement-oriented interactions aim to provide reasons and opportunities to continue. Together, these mechanisms establish a proposed cycle that supports both recovery after distraction and sustained participation in academic reading.

2.8 Synthesis of habit loop

Together, these theories establish habit formation as the overarching framework for the Reading Machine. Frictional feedback and calming interaction are explored as speculative supporting mechanisms that may facilitate the transition away from distraction. However, the project's primary investigation focuses on two interconnected challenges: reconstructing reading context after interruption and sustaining motivation once reading resumes. The former informs contextual recovery cues, while the latter informs the exploration of satisfying sensory interactions, achievable reading goals and interactive content presentation. By connecting these mechanisms within a repeatable habit loop, the research investigates how interaction design might support not only returning to academic reading but also making the experience more desirable to continue.

3 METHODOLOGY OF OVERALL PROJECT

This project adopts a practice-based approach informed by Research through Design, combining theoretical investigation with iterative design exploration. Literature informs the initial design direction, while observations from making and prototyping generate questions for further research and reflection. The process comprises four overlapping stages, allowing earlier assumptions and decisions to be revisited as the project develops.

Stage One — Research and Problem Framing
A focused literature review establishes the research problem, theoretical framework and scope. Habit formation provides the overarching framework, with task re-entry and sustained engagement identified as the primary research focuses. Frictional feedback and calming interaction are considered supporting, speculative mechanisms.

Stage Two — Ideation and Initial Prototyping
Structured ideation methods, including Crazy Eights and forced association, generate alternative design concepts. Sketches and exploratory prototypes translate these ideas into forms and interactions that can be critically examined. Concepts are compared according to their relevance to the research aims, interaction demands

and feasibility, informing the selection of directions for further development.

Stage Three — Development Through Practice
Selected directions are developed through precedent analysis, material experiments, drawing, digital modelling and interface exploration. These activities investigate sensory interaction qualities and reading-support functions. Observations are documented and critically interpreted alongside relevant literature, informing decisions to retain, modify or abandon particular approaches.

Stage Four — Synthesis and Final Proposition
Insights from research and practice are integrated into a more resolved design proposition. This is communicated through a partially functional prototype, supported by physical artefacts, interaction demonstrations and renders where appropriate. The outcome articulates how the proposed system might support returning to academic reading and motivation to continue, clearly distinguishing functional, simulated and speculative elements.

Evaluation relies on researcher-led critical reflection, considering alignment with the research aims, clarity of interaction and potential competition with reading for attention. Given time and ethics constraints, the project does not include formal user testing. Its contribution is therefore an exploratory design proposition and an account of the insights developed through practice, rather than validated evidence of effectiveness.

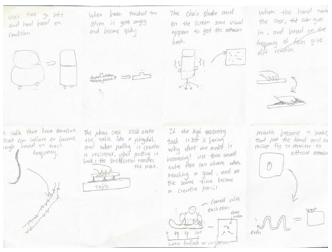
4 INITIAL IDEALISATION

To make the ideation process playful but still research-led, I used a controlled forced-association method combined with Crazy 8.I first selected common desk objects, then paired them with haptic reactions such as heartbeat, vibration, resistance, tension, warmth, or texture change. I then added an emotional condition, such as tiredness, anxiety, protectiveness, guilt, or calmness. The final step was to invent a metaphor that made the combination feel meaningful. For example, "phone case + heartbeat + tired" became the idea of a living phone skin that becomes exhausted when touched too often (see Figure 1).

This method helped me imagine the desktop as an emotional and responsive environment rather than a passive surface. In the forced-association process, A and B were compulsory elements, while C was optional and worked more as a reminder or hint. I then used the Crazy 8 method, where I had one minute to brainstorm and draw one idea for each random combination. Each page included eight quick ideas. I completed one Crazy 8 session per day over three days, rather than doing everything in one day, because I felt that taking breaks helped refresh my thinking and generate more creative ideas.

1	Object	Haptic Reaction	Feeling / condition
2	Phone Case	Vibrare	Calm
3	Charging Dock	Shake	Tired/Motivated
4	Mouse Pad	Breath	Playful/Boring
5	Mouse	Grip	Caring
6	Cable	Bounce	Proud
7	Desk Companion	Smooth / Rougher	Shy
8	Chair	Soft / Stiff	Satisfied

Figure 1. The table of words used for idea generation.



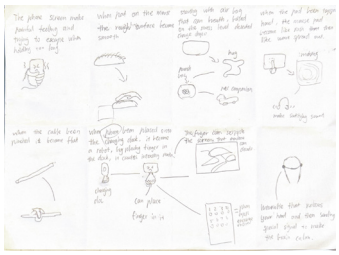


Figure 2 & 3. Rough sketches I drew.

3.1 Selected idea

Among the initial concepts, I was particularly interested in an airbag-inspired artefact that could change shape to support different interactions, functioning as a huggable object, a breathing companion or a small punching bag. This versatility connected to my interest in authenticity through the possibility of accommodating different users, preferences and emotional needs.

However, the mechanics required for continuous shape transformation exceeded the project's technical scope. I therefore narrowed the initial proposition to an air-filled mouse (see Figure 4). Although its implementation remained challenging, this form offered a more focused direction supported by existing product and research precedents.

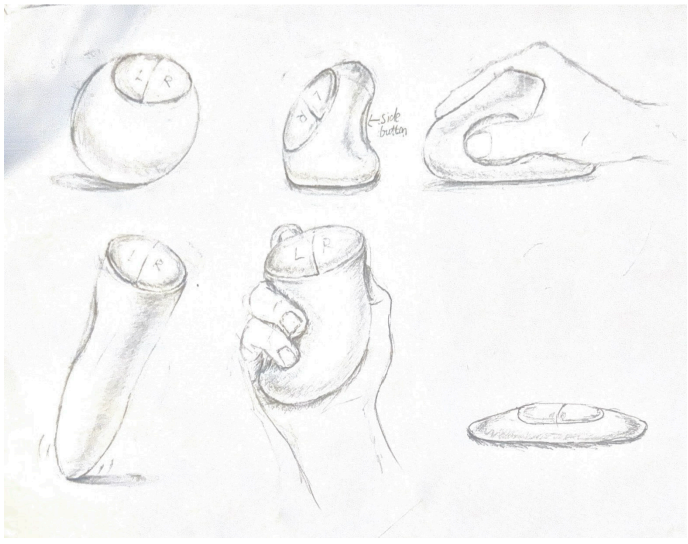


Figure 4. The mouse idea inspired by air bag

The mouse alone did not sufficiently address the wider reading-support system. Its proposed sensory interactions primarily related to frictional feedback and calming, leaving contextual task re-entry and motivation to continue reading underdeveloped. I therefore incorporated elements from a second concept to address these stages of the proposed habit loop.

The second concept proposed a handheld stick as an alternative input device, combining on-screen interaction with haptic feedback, such as vibrations corresponding to displayed content (see Figure 5). Inspired by the tendency to scribble when bored, it explored whether writing a sentence or question directly on screen could prompt an immediate AI response. The intention was to make the reading interface more responsive and playful, creating additional opportunities to interact with academic material.

The first design proposition combined the air mouse from the first concept with the screen-based interactions from the second. The mouse replaced the proposed stick, while retaining its approach to digital input and responsive reading assistance.

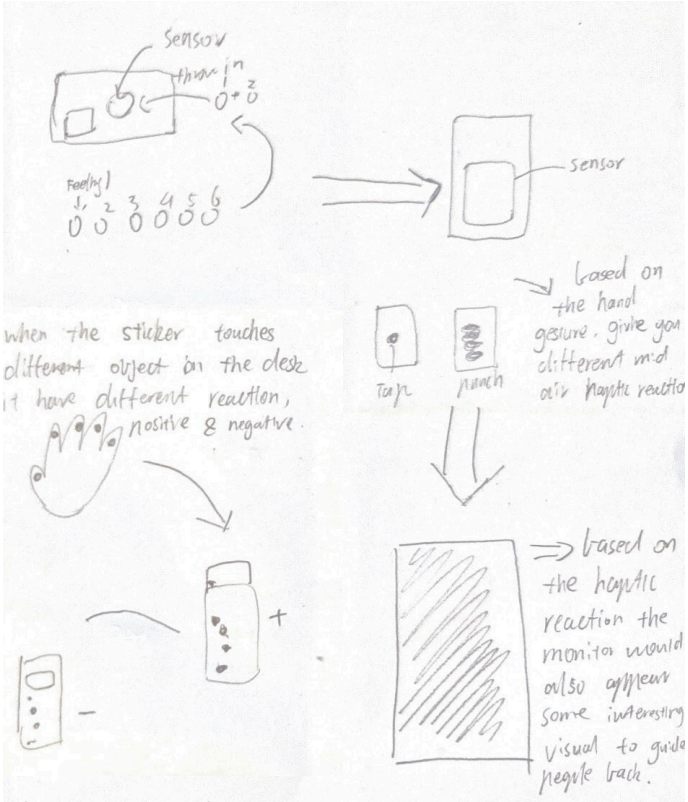


Figure 5. The sticker Idea

3.2 Developed first idea

The Air Mouse

The Air Mouse combines the functional role of a computer mouse with the deformable qualities of an air-filled object. Its proposed shape-changing structure accommodates computer navigation, handwritten input and tactile interaction. The side button provides access to the proposed controls: a single click activates custom shaping, holding restores the original form, a double click enables pressure adjustment, and a subsequent double click locks the modified shape. A triple click cycles between breathing, vibration and subtle feedback modes (see figure 6).

The proposed mechanism combines pressure sensing, AI-assisted control and a soft robotic structure to adjust the mouse's shape and resistance. During computer use, it would maintain a stable form intended to accommodate the user's grip. (see figure 7). During reading, thinking or brief pauses, it could be held and squeezed as a tactile object. This combination explores how sensory interaction might be incorporated into an existing study tool and remain accessible within the reading workflow.

Different pressure settings would provide contrasting interaction qualities. A squeezing mode would offer gradual resistance and recovery, while a softer mode would accommodate stronger compression or tapping (see figure 8). A side button would allow users to switch between settings. These variations are intended to accommodate different preferences and emotional needs, although their suitability would require further investigation.

The Air Mouse also proposes a breathing mode (figure 9), using slow cycles of expansion and contraction to provide a tactile rhythm that users could follow during a brief pause which mainly serve the idea of calming user down after been distracted. Adjustable pressure and movement settings would allow users to modify the feedback. At the same time it can slightly move left to right to create friction when the user has been distracted.

Alongside these physical interactions, a pen mode would allow users to annotate reading materials and write questions directly within the digital interface (see figure 10). For example, writing "What does this mean?" beside a passage would prompt an AI-generated explanation. Vibration would provide feedback during input, although the mechanism connecting the device's movement to on-screen writing would require further technical development.

Integrating annotation and assistance within the reading interface aims to reduce switching between separate tools and make engagement with academic material more immediate and interactive. Whether these functions support comprehension, task re-entry or motivation to continue remains a question for subsequent investigation.

Together, these proposed functions position the Air Mouse as both an input device and a sensory component within a reading-support system. The initial design provides a starting point for investigating how physical interaction and digital feedback might jointly support academic reading.



Figure 6. Side view of air mouse with all buttons.



Figure 7. Air mouse in ergonomic mouse form.



Figure 8. Air mouse when breathing and shaking.

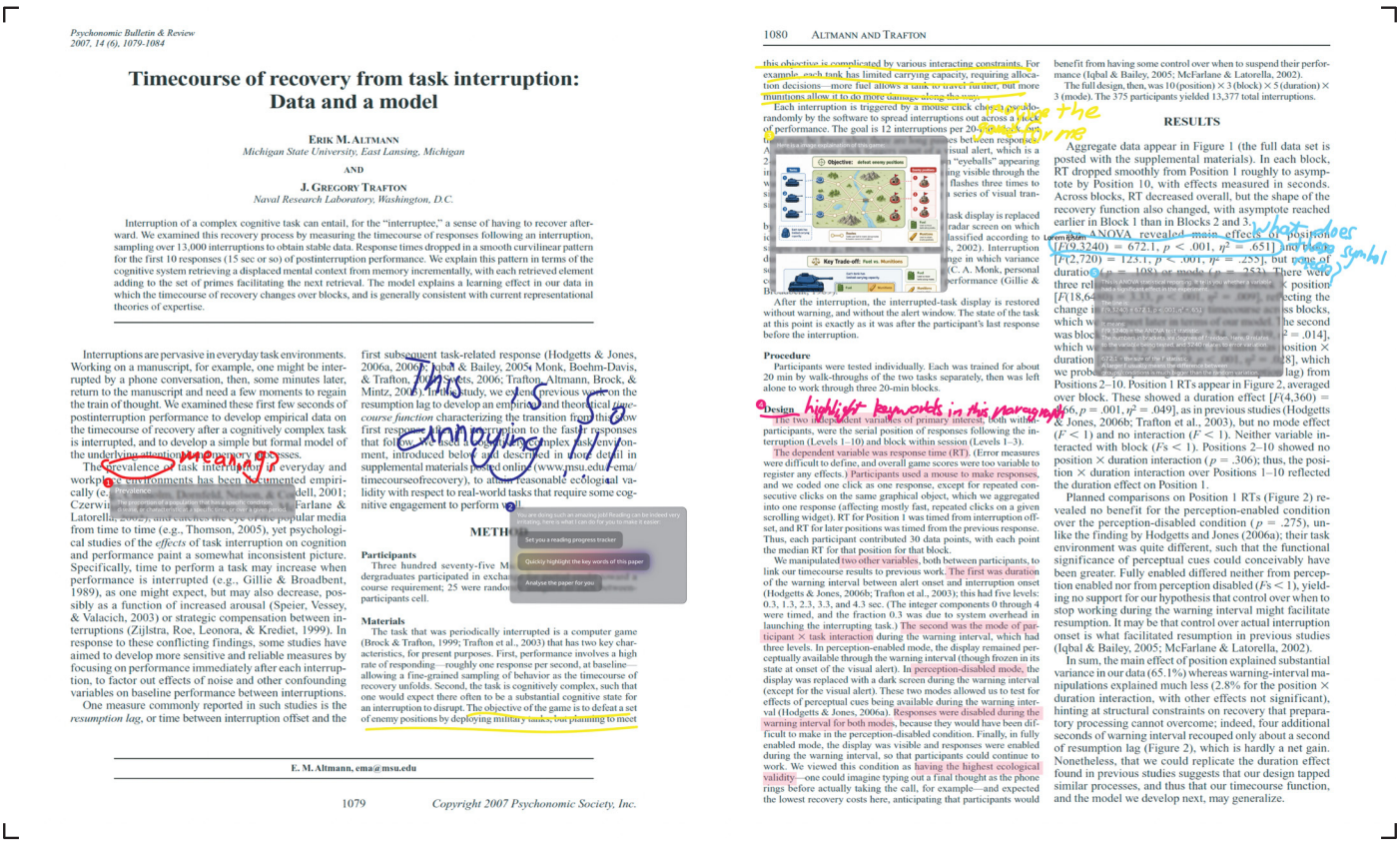


Figure 9. Air mouse in stress reliefing mode.



Figure 10. Air mouse in scribble mode.

On Screen System



Psychonomic Bulletin & Review
2007, 14 (6), 1079-1084

Timecourse of recovery from task interruption: Data and a model

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Interruption of a complex cognitive task can entail, for the "interruptee," a sense of having to recover afterward. We examined this recovery process by measuring the timecourse of responses following an interruption, sampling over 13,000 interruptions to obtain stable data. Response times dropped in a smooth curvilinear pattern for the first 10 responses (15 sec or so) of postinterruption performance. We explain this pattern in terms of the cognitive system retrieving a displaced mental context from memory incrementally, with each retrieved element adding to the set of primes facilitating the next retrieval. The model explains a learning effect in our data in which the timecourse of recovery changes over blocks, and is generally consistent with current representational theories of expertise.

Interruptions are pervasive in everyday task environments. Working on a manuscript, for example, one might be interrupted by a phone conversation, then, some minutes later, return to the manuscript and need a few moments to regain the train of thought. We examined these first few seconds of postinterruption performance to develop empirical data on the timecourse of recovery after a cognitively complex task is interrupted, and to develop a simple but formal model of the underlying attention and memory processes.

The prevalence of task interruption in everyday and workplace environments has been documented empirically (e.g., Gillie & Broadbent, 1989; Hodgetts & Jones, 2006a, 2006b; Kirmeyer, 1988; Laiddell, 2001; Czerwinski, Horvitz, & Wilhite, 2004; Farlane & Latorella, 2002), and catches the eye of the popular media from time to time (e.g., Thomson, 2005), yet psychological studies of the *effects* of task interruption on cognition and performance paint a somewhat inconsistent picture. Specifically, time to perform a task may increase when performance is interrupted (e.g., Gillie & Broadbent, 1989), as one might expect, but may also decrease, possibly as a function of increased arousal (Speier, Vessey, & Valacich, 2003) or strategic compensation between interruptions (Zijlstra, Roe, Leonora, & Krediet, 1999). In response to these conflicting findings, some studies have aimed to develop more sensitive and reliable measures by focusing on performance immediately after each interruption, to factor out effects of noise and other confounding variables on baseline performance between interruptions.

One measure commonly reported in such studies is the *resumption lag*, or time between interruption offset and the first subsequent task-related response (Hodgetts & Jones, 2006a, 2006b; Iqbal & Bailey, 2005; Monk, Boehm-Davis, & Trafton, 2004; Trafton, 2006; Trafton, Altmann, Brock, & Mintz, 2003). In this study, we extend previous work on the resumption lag to develop an empirical and theoretical *timecourse function* characterizing the transition from the interruption to the first response. The interruption to the fast responses that follow. We tested a complex task environment, introduced below and described in more detail in supplemental materials posted online (www.msu.edu/~ema/timecourseofrecovery), to attain reasonable ecological validity with respect to real-world tasks that require some cognitive engagement to perform well.

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this objective is complicated by various interacting constraints. For example, each tank has limited carrying capacity, requiring allocation decisions—more fuel allows a tank to travel further, but more munitions allow it to do more damage along the way.

Each interruption is triggered by a mouse click (keypad pseudorandomly by the software to spread interruptions out across a block of performance. The goal is 12 interruptions per 20-min block, with

After the interruption, the interrupted-task display is restored without warning, and without the alert window. The state of the task at this point is exactly as it was after the participant's last response before the interruption.

Procedure

Participants were tested individually. Each was trained for about 20 min by walk-throughs of the two tasks separately, then was left alone to work through three 20-min blocks.

Design

The two independent variables of primary interest, both within participants, were the serial position of responses following the interruption (Levels 1–10) and block within session (Levels 1–3).

The dependent variable was response time (RT). (Error measures were difficult to define, and overall game scores were too variable to register any effects.) Participants used a mouse to make responses, and we coded one click as one response, except for repeated consecutive clicks on the same graphical object, which we aggregated into one response (affecting mostly fast, repeated clicks on a given scrolling widget). RT for Position 1 was timed from interruption offset, and RT for later positions was timed from the previous response. Thus, each participant contributed 30 data points, with each point the median RT for that position for that block.

We manipulated two other variables, both between participants, to link our timecourse results to previous work. The first was duration of the warning interval between alert onset and interruption onset (Hodgetts & Jones, 2006b; Trafton et al., 2003); this had five levels: 0.3, 1.3, 2.3, 3.3, and 4.3 sec. (The integer components 0 through 4 were timed, and the fraction 0.3 was due to system overhead in launching the interrupting task.) The second was the mode of participant × task interaction during the warning interval, which had three levels. In perception-enabled mode, the display remained perceptually available through the warning interval (though frozen in its state at onset of the visual alert). In perception-disabled mode, the display was replaced with a dark screen during the warning interval (except for the visual alert). These two modes allowed us to test for effects of perceptual cues being available during the warning interval (Hodgetts & Jones, 2006a). Responses were disabled during the warning interval for both modes, because they would have been difficult to make in the perception-disabled condition. Finally, in fully enabled mode, the display was visible and responses were enabled during the warning interval, so that participants could continue to work. We viewed this condition as having the highest ecological validity—one could imagine typing out a final thought as the phone rings before actually taking the call, for example—and expected the lowest recovery costs here, anticipating that participants would benefit from having some control over when to suspend their performance (Iqbal & Bailey, 2005; McFarlane & Latorella, 2002).

The full design, then, was 10 (position) × 3 (block) × 5 (duration) × 3 (mode). The 375 participants yielded 13,377 total interruptions.

RESULTS

Aggregate data appear in Figure 1 (the full data set is posted with the supplemental materials). In each block, RT dropped smoothly from Position 1 roughly to asymptote by Position 10, with effects measured in seconds. Across blocks, RT decreased overall, but the shape of the recovery function also changed, with asymptote reached earlier in Block 1 than in Blocks 2 and 3.

A three-way ANOVA revealed main effects of position ($F(9,3240) = 672.1$, $p < .001$, $\eta^2 = .651$) and block ($F(2,720) = 123.1$, $p < .001$, $\eta^2 = .255$), but none of duration ($p = .108$) or mode ($p = .253$). There were three rel… position ($F(18,64…$ ecting the change i… ss blocks, which w… he second was bloc… $= .014$], which w… osition × duration … 28], which we prob… lag) from Positions 2–10. Position 1 RTs appear in Figure 2, averaged over block. These showed a duration effect [$F(4,360) = 3.66$, $p = .001$, $\eta^2 = .049$], as in previous studies (Hodgetts & Jones, 2006b; Trafton et al., 2003), but no mode effect ($F < 1$) and no interaction ($F < 1$). Neither variable interacted with block ($Fs < 1$). Positions 2–10 showed no position × duration interaction ($p = .306$); thus, the position × duration interaction over Positions 1–10 reflected the duration effect on Position 1.

Planned comparisons on Position 1 RTs (Figure 2) revealed no benefit for the perception-enabled condition over the perception-disabled condition ($p = .275$), unlike the finding by Hodgetts and Jones (2006a); their task environment was quite different, such that the functional significance of perceptual cues could conceivably have been greater. Fully enabled differed neither from perception enabled nor from perception disabled ($Fs < 1$), yielding no support for our hypothesis that control over when to stop working during the warning interval might facilitate resumption. It may be that control over actual interruption onset is what facilitated resumption in previous studies (Iqbal & Bailey, 2005; McFarlane & Latorella, 2002).

In sum, the main effect of position explained substantial variance in our data (65.1%) whereas warning-interval manipulations explained much less (2.8% for the position × duration interaction, with other effects not significant), hinting at structural constraints on recovery that preparatory processing cannot overcome; indeed, four additional seconds of warning interval recouped only about a second of resumption lag (Figure 2), which is hardly a net gain. Nonetheless, that we could replicate the duration effect found in previous studies suggests that our design tapped similar processes, and thus that our timecourse function, and the model we develop next, may generalize.

Figure 11. Design on On Screen System

- Find definiton function
- When the user is stressful, the AI read its emotion and help them choosing a task, so learners can boost their motivation by establishing a clearly achievable goal (Hui, 2025)
- There are many times when users may feel frustrated by complex descriptions or confusing information that is difficult to understand. In these situations, AI can help by quickly visualising the content, making it easier for users to understand and continue their work.
- Highlighting can break a complex paragraph into smaller and easier sub-tasks. More achievable goal gives the user more motivation to finish.
- Users can also use scribbling to decode complex questions, making the reading flow smoother without needing to stop, search separately, or overthink the meaning.

3.3 Critsism and Direction

Desirability and Reading Support

The initial proposition offered a potentially engaging physical interaction through the Air Mouse, but its digital component provided limited support for motivation to continue reading. The proposed AI assistance largely reproduced existing approaches, leaving the distinctive value of the reading experience insufficiently developed. Task re-entry cues were also underdeveloped, particularly those intended to help readers recover their previous position, understanding and next action after interruption. Furthermore, the design lacked a defined mechanism for recognising distraction or allowing users to signal that they needed support, leaving the trigger for the proposed recovery sequence unclear. Although the mouse's shape-changing capability remained speculative, its proposed sensory qualities could be investigated through material experiments.

Integration of Physical and Digital Interaction

The physical and digital components operated largely independently, weakening the rationale for presenting them as an integrated product system. The Air Mouse functioned as an alternative input device, while the interface offered reading assistance with little connection to its sensory qualities. A more coherent relationship could emerge through corresponding physical and digital behaviours, such as translating squeezing and release into visible deformation and recovery on screen.

Directions for Further Exploration

Further exploration would investigate squishy materials to inform the Air Mouse's shape-changing behaviour, pressure and material qualities; examine how squishy interactions could translate into engaging and potentially stress-relieving digital experiences; and explore how digital cues might support task re-entry and self-directed motivation to continue reading. And overall, this is an design project, besides the mechanics, appeareance and design style also need to be invesigated

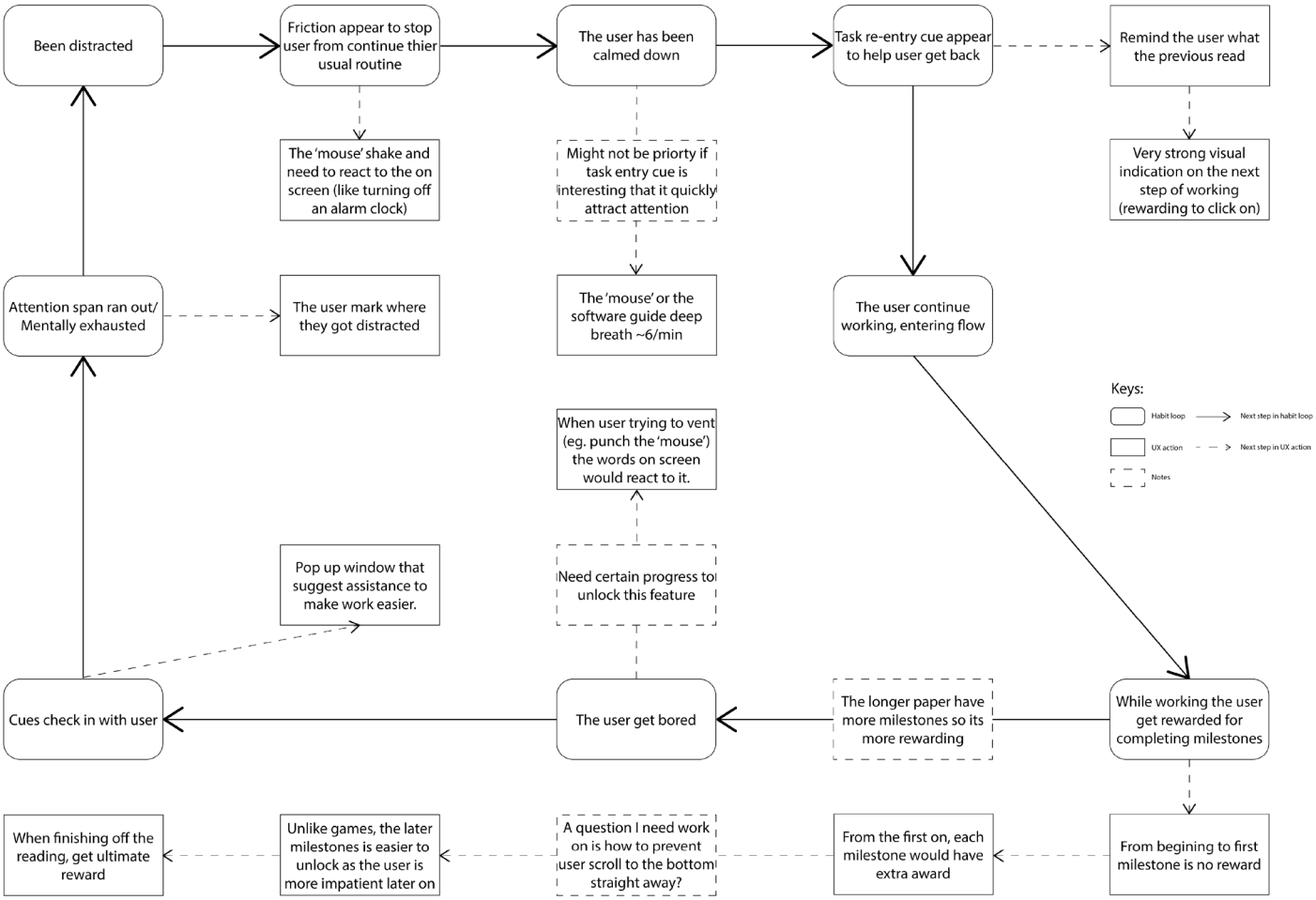


Figure 12. Proposed reading habit loop identifying opportunities for physical and digital interventions to support task re-entry and sustained engagement.

4 PRECEDENT REVIEW

One key things in the inital idea is that I did do insufdfience knowledge on exsting reading machines thus leading to a very unploished design that mostly adresses the exsring designs. So this research mainly for looking into precdent find the unoque positioning of my porject both mechanically and visually.

4.1 Similar reading machines Research

I conducted a comparative feature analysis of Acrobat, WPS PDF, Kindle, Apple Books and Readwise Reader to identify established reading functions and opportunities for further design exploration (see Figure 13). The review grouped functions into three layers according to their proposed role within the system. This classification informed design priorities rather than evaluating the overall quality of each product.

Layer One — Core Reading Functions

Navigation, layout controls, search, bookmarks, highlighting and annotation establish the functional foundation of the reading experience. Familiar behaviours such as scrolling and adjusting text size would largely be retained, allowing development to concentrate on interactions more directly relevant to the research aims. Selected functions, including highlighting and position marking, offered opportunities to explore satisfying feedback without unnecessarily complicating reading.

Layer Two — Reading Assistance

Definitions, translation, text-to-speech and AI-generated explanations or summaries provide supplementary support. However, adding these capabilities alone would not address the initial proposition's limited emphasis on motivation to continue reading. They were therefore assigned a supporting role and a lower priority for practical exploration.

Layer Three — Task Re-entry and Sustained Engagement

The third layer identifies opportunities most closely aligned with the research question, including progressive content reveal, reading guidance, contextual recovery cues, retrieval-based interaction and responsive typography. Selected precedents demonstrated related approaches, such as line guides and review activities. These informed opportunities to adapt and combine existing mechanisms rather than treating every proposed interaction as entirely new.

The analysis established a hierarchy for subsequent development: retain essential functionality, incorporate assistance where relevant, and prioritise experiments connecting task re-entry with motivation to continue. It

also provided a basis for examining whether sensory and interactive feedback could serve a meaningful reading function rather than operate as an additional distraction. Some of the specific funtion that anaylsed by me as the design inspiration has been put in appendix 1.

4.2 Interactive Cues Research

The precedent review identified six interaction directions for further investigation: content segmentation, paced highlighting, attention-capture cues, retrieval-based interaction, AI-supported concept linking and expressive typography. These address two connected priorities within the proposed habit loop: helping readers recover their context after distraction and supporting motivation to continue reading. Research on segmentation, progress monitoring and signaling provides the primary evidence informing these explorations.

Content Segmentation and Visible Progress

The progressive-reveal precedents suggested ways to divide long texts into shorter reading sections, displaying the current section prominently while concealing or minimising surrounding content (see Figure 13). This informed the proposition that readers could encounter a

Reading Functions	Acrobat	WPS PDF	Kindle	Apple Books	Readwise Reader
Page / scroll navigation	✓	✓	✓	✓	✓
Zoom / text-size control	✓	✓	✓	✓	✓
Reading layout / theme	◐	✓	✓	✓	✓
Table of contents / navigation	✓	✓	✓	✓	✓
Search within text	✓	✓	✓	✓	✓
Bookmark / save position	✓	✓	✓	✓	✓
Select & copy text	✓	✓	◐	✓	✓
Highlight	✓	✓	✓	✓	✓
Notes / annotation	✓	✓	✓	✓	✓
Reading ruler / line guide	–	–	✓	✓	–
Translation	◐	✓	◐	✓	✓
Read aloud / TTS	✓	✓	✓	◐	✓
Dictionary / quick definition	◐	◐	✓	✓	✓
AI summary	✓	✓	–	–	–
AI question / explanation	✓	✓	◐	–	–
Related-text exploration	–	–	◐	–	◐
Chapter unlock	–	–	–	–	–
Progressive section reveal	–	–	–	–	◐
Paced moving highlight	–	–	◐	◐	–
Interactive typography	–	–	–	–	–
Attention-capture interaction	–	–	–	–	–
Active task-re-entry cue	–	–	–	–	◐

Key

✓ = Established feature

◐ = Related/partial implementation

– = Not identified

= Layer 1 that will mostly adapt

= Layer 1 that will futher explore

= Layer 2

= Layer 3

Figure 12. Table for analysing major reading machines on the market

manageable immediate task while retaining an overview of the wider text.

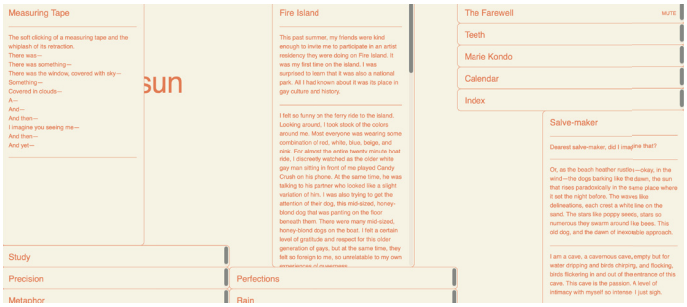


Figure 13. Precedents for organising text into sections and changing the prominence of selected content.

Rey et al. (2019, p. 389) describe the segmenting effect in terms of presenting instruction in "meaningful and coherent" segments. Their meta-analysis found benefits for retention and transfer, reduced cognitive load and increased learning time. These findings informed the exploration of section-based reading and learner-controlled progression. However, the research concerns multimedia instruction; expanding the current text section and minimising surrounding sections represent design applications to investigate rather than independently validated techniques.

Progress indicators and milestones could complement this structure by making completed and remaining reading visible. Harkin et al. (2016, p. 198) conclude that "monitoring goal progress is an effective self-regulation strategy." Their meta-analysis found that interventions encouraging progress monitoring promoted goal attainment. This supports exploring reading progress bars, section indicators and proximity to the next goal.

Paced Highlighting and Reading Guidance
The moving-highlight precedents suggested a visual guide that readers could follow through a passage (see Figure 12). This could help maintain reading position and provide a sense of forward movement during longer texts.



Figure 13. Typographic and highlighting precedents.

Schneider et al. (2018, p. 1) describe signaling as cues that "highlight the organization of specific relevant information within materials." Their meta-analysis found positive effects on retention and transfer, supporting the use of purposeful visual guidance. This informed the exploration of highlighted text regions, reading rulers, key-term emphasis and directional cues.

Nevertheless, signaling evidence does not demonstrate that a fixed moving highlight suits every reader. Reading speed varies with familiarity, sentence complexity and the need to reflect. The proposed interaction would therefore explore adjustable or semi-controlled pacing, allowing readers to pause, slow down or resume rather than follow an imposed speed.

Attention Capture and Contextual Task Re-entry
The target-catching precedents suggested brief interactions that could redirect attention towards the interface after distraction (see Figure 14). The attraction of these examples lay in their combination of movement, immediate response and a clear action. However, capturing attention alone would not restore the reader's previous understanding.

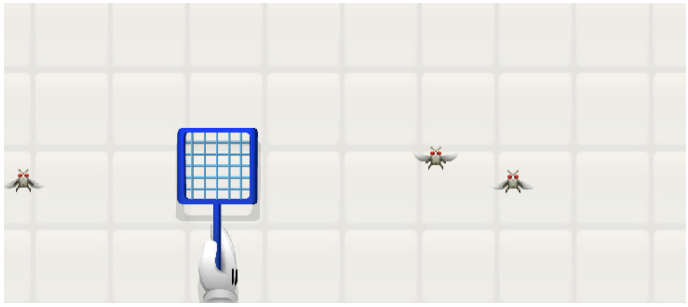


Figure 14. Target-based interaction precedents informing the exploration of attention-capture cues.

The proposed sequence therefore connects attention capture with contextual guidance: a brief responsive interaction would lead directly to a highlighted reading position, a reminder of previous content or an identifiable next action. Schneider et al. (2018) provide a basis for making relevant information visually prominent, while Altmann and Trafton's (2007) account of incremental contextual recovery explains why re-entry requires more than looking back at the screen. The specific moving-target mechanism remains exploratory and would need to avoid becoming a separate distraction.

Retrieval-Based Interaction and Concept Linking
Sentence reconstruction and concealed-information precedents offered ways to revisit previously read material (see Figure 15). A selected sentence could be reconstructed, or a short prompt could invite recall before revealing the original content. These interactions could support contextual recovery after interruption or provide a reflective transition between reading sections.

However, reconstructing every word would be poorly suited to complex academic passages and could introduce unnecessary effort. Further exploration would therefore focus on short phrases, key concepts or selected information. Schneegass et al. (2022) provide supplementary support for investigating interactive

memory cues, while their findings distinguish perceived helpfulness from demonstrated performance improvements.

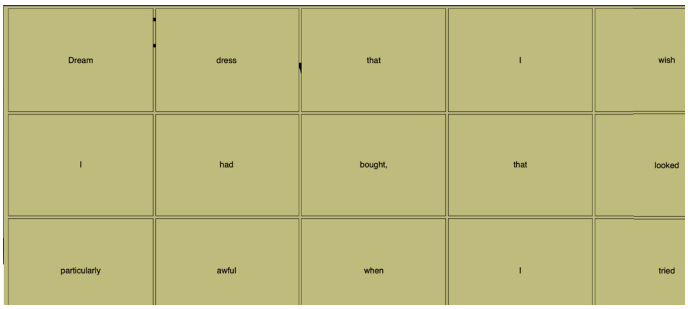


Figure 15. Sentence-reconstruction and concealed-information precedents

AI-supported concept linking offered a related opportunity to explore connections between selected terms, passages or arguments (see Figure 16). This could extend reading beyond passive presentation, but generated relationships would need to remain transparent and open to correction. Its contribution to motivation and understanding would remain a question for subsequent practice.

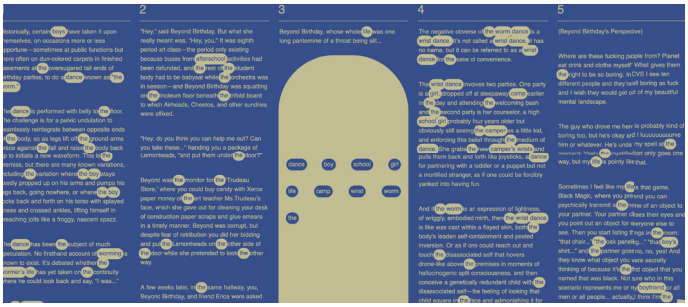


Figure 16. Concept-linking precedents showing relationships between selected words and passages.

Expressive Typography and Sensory Feedback
The responsive and liquid-like typography precedents demonstrated how text could become an interactive object through movement and deformation (see Figure 17). These qualities suggested possibilities for completion feedback, transitions and brief sensory interactions.



Figure 17. Responsive typography precedents informing deformation effects and sensory feedback.

Their role would remain secondary to the reading-support mechanisms above. Expressive effects might acknowledge progress, but neither progress-monitoring nor signaling research establishes that decorative rewards improve reading motivation. They would therefore be explored selectively, with attention to whether they reinforce a meaningful action or compete with the text.

Overall, segmentation, visible progress and reading guidance primarily inform sustained engagement, while attention capture and contextual retrieval primarily inform task re-entry. Some interactions span both purposes. Together, these directions establish a basis for investigating how the system could help readers return, identify a manageable next action and choose to continue.

The exploration will be prioritised according to the order of the cues that has been written.

4.3 Design Style Research

The visual precedent analysis explored how a playful, materially expressive style could inform the digital reading environment. Across the selected references, recurring characteristics included textured three-dimensional objects, inflated typography, irregular compositions and controlled colour palettes (see Figure 18). These observations informed three principles for subsequent design development.



Figure 18. Selected visual precedents informing the proposed interface style.

Material Expression
Metallic, foamy, rubber-like and plastic-like surfaces gave digital objects recognisable material qualities. Their contrast with flat backgrounds and typography made individual elements visually distinctive. This suggested that digital squishies should be differentiated through surface qualities as well as shape and movement. Inflated typography offered a related direction, allowing selected words to share the material language of interactive objects. These treatments would be reserved for headings, feedback and sensory interactions rather than extended reading.

Expressive Composition and Reading Structure
Although many references contained visually complex elements, these were often balanced by simple backgrounds and generous negative space. Cut-out imagery, irregular containers and variations in scale reduced the visibility of the underlying grid without removing its organisational role.

For the reading interface, this suggested retaining a clear structure while allowing selected interactive elements to extend beyond conventional rectangular containers. Visual expression would be concentrated around purposeful interactions, while secondary controls could remain concealed until needed. This approach aimed to preserve space for reading without losing the playful character identified in the references.

Colour, Typography and Surface Detail
The references frequently used expressive colours with moderated saturation and shared warm undertones. This informed the development of a cohesive palette that could accommodate variation without making every element equally prominent. Subtle grain also offered a way to introduce surface detail and connect otherwise contrasting visual elements.

Mixed typefaces contributed personality, but required a consistent hierarchy. A conventional typeface would therefore support the main reading text, with decorative styles used selectively for titles, transitions and interactive feedback. Together, these findings established a visual direction combining expressive material details with a restrained reading surface, to be refined through subsequent interface experiments.

The full anylsis of the designs style research has been placed in appendix 2, this section keep the keep findings

5 EXPEREIMENTS

Physical experimentation formed a central part of the practice-based exploration, enabling direct investigation of sensory qualities that could not be adequately understood through digital references alone. Hands-on engagement with materials informed design decisions and generated new directions for subsequent exploration.

5.1 Squishy Testing

Squishy interactions form a central part of the proposed user experience, both physically through the Air Mouse and digitally through responsive interface elements. I therefore conducted an exploratory material comparison to investigate which sensory qualities I found satisfying under different interaction conditions and how these might inform subsequent design development.

The sample included squishies with gel or hydrogel fillings, thermoplastic rubber (TPR) structures and polyurethane (PU) foam. These categories were used as a practical basis for comparison rather than a verified chemical classification; for example, a gel-filled object may also have a TPR outer skin. Where products within a category offered noticeably different softness or resistance, I included contrasting examples. Melamine foam, play dough and slime were also included as comparative materials to help distinguish qualities such as compressibility, deformation and recovery.

Sample selection prioritised rounded objects that could sit within the palm and be manipulated with one hand. This reduced variation in handling and reflected the intended scale of the Air Mouse. Objects that did not meet these criteria were excluded and documented. One foam sample was smaller than intended because a suitable alternative could not be sourced. Selection also aimed to represent commonly encountered qualities within each material category, while retaining contrasting samples where their sensory behaviour differed substantially. Odour was excluded as an evaluation criterion because the investigation focused on touch, appearance and sound.

Physical testing involved six actions: touching, pressing, pinching, squeezing, punching downwards and repeated squeezing for 30 seconds. I aimed to keep the duration, number of repetitions and applied effort consistent across samples, although force was not instrumentally measured. Each action was documented through photographs and reflective notes describing my sensory impressions. Between tests, I typed freely for one minute to introduce a consistent interval and approximate movement between tactile interaction and computer-based work. This interval was not treated as a verified sensory reset.

The comparison generated situated observations to inform the Air Mouse's proposed material qualities and the translation of physical behaviour into digital interaction. It did not assess stress reduction or establish preferences representative of other users.

The experiment was documented through three complementary visual records. The first table presents photographs and qualitative observations of each material across the tested actions (Figure 19). The radar charts then compare their perceived sensory

qualities and outline a preferred profile to guide further exploration (Figure 20). Finally, the heatmap ranks my perceived comfort and suitability of each material for different actions, informing the sensory behaviours the Air Mouse could aim to reproduce (Figure 21).

Results

Across the six materials, the strongest-performing squishies share a relatively consistent sensory profile. High deformation softness, strong cushioning and low surface friction appear to be the most favourable characteristics. Materials that feel soft while being deformed and absorb pressure without reaching a hard stopping point performed better across the interaction tasks. Low surface friction also appears important because it allows the hand to move, press and repeatedly manipulate the object with less drag and physical effort.

In contrast, characteristics such as compression resistance, elastic rebound, recovery speed and force/ speed sensitivity appear to work best at a moderate level rather than at either extreme. Too much resistance can

Squishies Sensory Response Experiement

Types of Squishy / Types of Haptic Action	Soft PU	Hard PU	TPR - Water Filling	TPR – Hydrogel Filling	Melamine foam	Play dough
Touch	The surface felt very smooth and slightly moist. The moisture is just enough for my finger to -glide completely freely across it.	The surface felt smooth but noticeably tacky. This created relatively high friction when sliding a finger across it, making it feel more resistant than Soft PU.	The outer TPR surface felt slightly sticky and wet, but the soft water filling underneath made the overall contact feel flexible and soft. The finger could still move across the surface relatively easily.	The outer TPR surface felt slightly rough and wet. Compared with the water-filled version, the denser gel underneath made the object feel firmer and more substantial when touched.	The surface felt dry and finely textured. Although it was not strongly uneven, it had a noticeably rough, almost fine-sandpaper-like quality.	The surface felt smooth but highly tacky. My finger could move across it, but the stickiness created noticeable drag.
Press	Very little force was required to press the material down. At full compression, it spread softly around the fingers, creating an enveloping or "hugging" sensation rather than a strong push-back.	Pressing required noticeably more force. The material resisted compression and produced a springy push-back, creating a firmer and more elastic response than Soft PU.	The material was very soft but at the same time bouncy. So when I press, a not so strong movement, it instantly bounce my finger back with. Resistance through my pressing experience has been felt.	The response depended strongly on how the pressure was applied. Slow pressing allowed the gel to deform gradually and feel relatively soft, while faster or stronger pressing created much greater resistance and could make the material feel temporarily very firm.	The foam compressed very easily and offered little progressive resistance. The finger through did feel slight resistance, so the interaction felt a lot harder than norman pu and could not feel the gradual cushion.	The material provided steady resistance throughout the press. At deeper compression the resistance became strong enough to create noticeable pressure on the fingertip, making prolonged pressing less comfortable.
Pinch	Pinching occurred in two stages. Initial compression required very little effort, followed by gentle resistance as the material folded together. Even at full compression, a soft buffer remained between the fingers.	Pinching produced a springier and more elastic response. The material resisted compression and quickly pushed back, so the sensation felt more bouncy than deeply cushioned.	The water shifted easily away from the fingers, allowing the TPR skin to stretch and fold. The interaction felt soft, fluid and elastic, with relatively little internal resistance. Less elasticity than press action.	A single fast or forceful squish could feel uncomfortable because the dense gel strongly resisted sudden deformation. However, slower squeezing allowed the gel to move and deform more gradually, producing a much softer response. Feeling the gel to deform is satisfying	The foam itself felt soft, but once fully compressed there was limited cushioning remaining between the fingers. This made stronger pinching less comfortable.	The material became very soft during pinching and responded slowly. It provided consistent resistance throughout the movement while still leaving a soft cushion between the fingers at full compression.
Squish	The material was extremely soft and required very little effort to deform. The first squish felt highly satisfying, but it reached full compression quickly, giving only a small range of changing resistance.	The material produced a much more energetic and elastic response. It changed shape quickly, resisted the hand more strongly and recovered with a noticeable bounce.	The water moved freely through the object, making it very soft, fluid and highly bouncy. The TPR skin stretched around the hand and rebounded quickly after release, creating a lively and playful squishing experience as watching it stretch and change shape is satisfying	Sudden impact produced very high resistance. Because the dense gel could not move away from the impact as freely as water, the object became temporarily very firm and transferred more force back into the hand, making the action uncomfortable.	The foam compressed and recovered almost immediately. Because the recovery happened so quickly, there was little time to observe or anticipate the return, which made the interaction feel less satisfying to me.	The material deformed slowly and visibly during the squish. Watching it change shape and move between the fingers was satisfying, although it produced little bounce or elastic return.
Punch down	Punching felt highly cushioned. The material offered little initial resistance but spread around the hand at full compression, helping prevent hard contact with the surface underneath. However it is little bit too soft that if I punch down hard, there is slightly short in cusioning, though overall soft expereince.	The material absorbed the impact while also producing a noticeable elastic response. Its springiness created a cushioned. Far more energetic sensation than Soft PU thus more comfortable.	The moving water helped redistribute the impact while the elastic TPR skin stretched and rebounded. Although the material provided relatively little cushioning, this was partly compensated by its strong bounce-back response, which pushed my hand away from the surface without causing discomfort.	The moving water helped redistribute the impact while the elastic TPR skin stretched and rebounded. Although the material provided relatively little cushioning, this was partly compensated by its strong bounce-back response, which pushed my hand away from the surface without causing discomfort.	The foam compressed immediately and offered little resistance, so the hand approached the hard table surface quickly. It provided some cushioning but not enough for stronger impact.	The material felt unexpectedly firm under fast impact. It absorbed some force but provided limited cushioning, making it more comfortable than the dense gel but less protective than the PU materials.
Continuously squish for 30 secs	After being fully compressed, the material took a relatively long time to return to its original shape. This slow recovery was visually satisfying. However under stress condition where I repeated squeezing before full recovery made the object feel denser and harder to compress. It therefore suited slower, occasional interaction better than continuous use.	The material recovered much faster and maintained a springy, elastic response. Its quicker rebound made repeated squeezing easier and more suitable for continuous interaction. However the rebound is around 3 seconds, still not fast enough for fast pace continuous squeeze.	The combination of freely moving water and elastic TPR allowed continuous squeezing with super effortless. The object stayed soft, fluid and bouncy, making repeated interaction feel playful and energetic. The bounce back is almost instant	This material became much more comfortable during repeated, slower manipulation than during a single forceful squish. Slow or continuous squeezing gave the gel time to move and deform, creating a softer response. Fast or aggressive compression still produced much stronger resistance.	Repeated squeezing began to feel more like hand exercise than stress relief. Although the material remained soft, I can still feel a little resistance which make experience less smooth. Its limited cushioning and immediate recovery created fatigue without providing much continuing sensory reward.	Continuous manipulation remained soft but gradually became physically tiring. Because the material had little elastic rebound and I also have to reshape it myself, the interaction felt more like kneading or exercising the hand than receiving repetitive stress-relieving feedback.

Figure 19. Table that records my observation

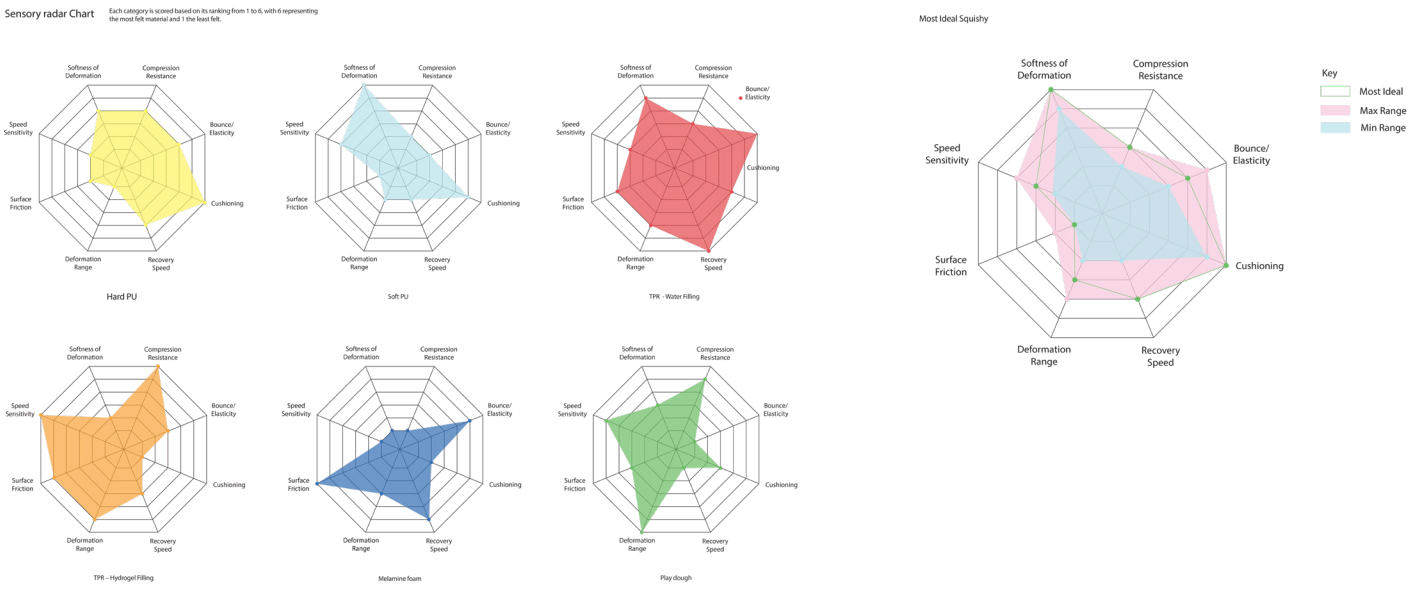
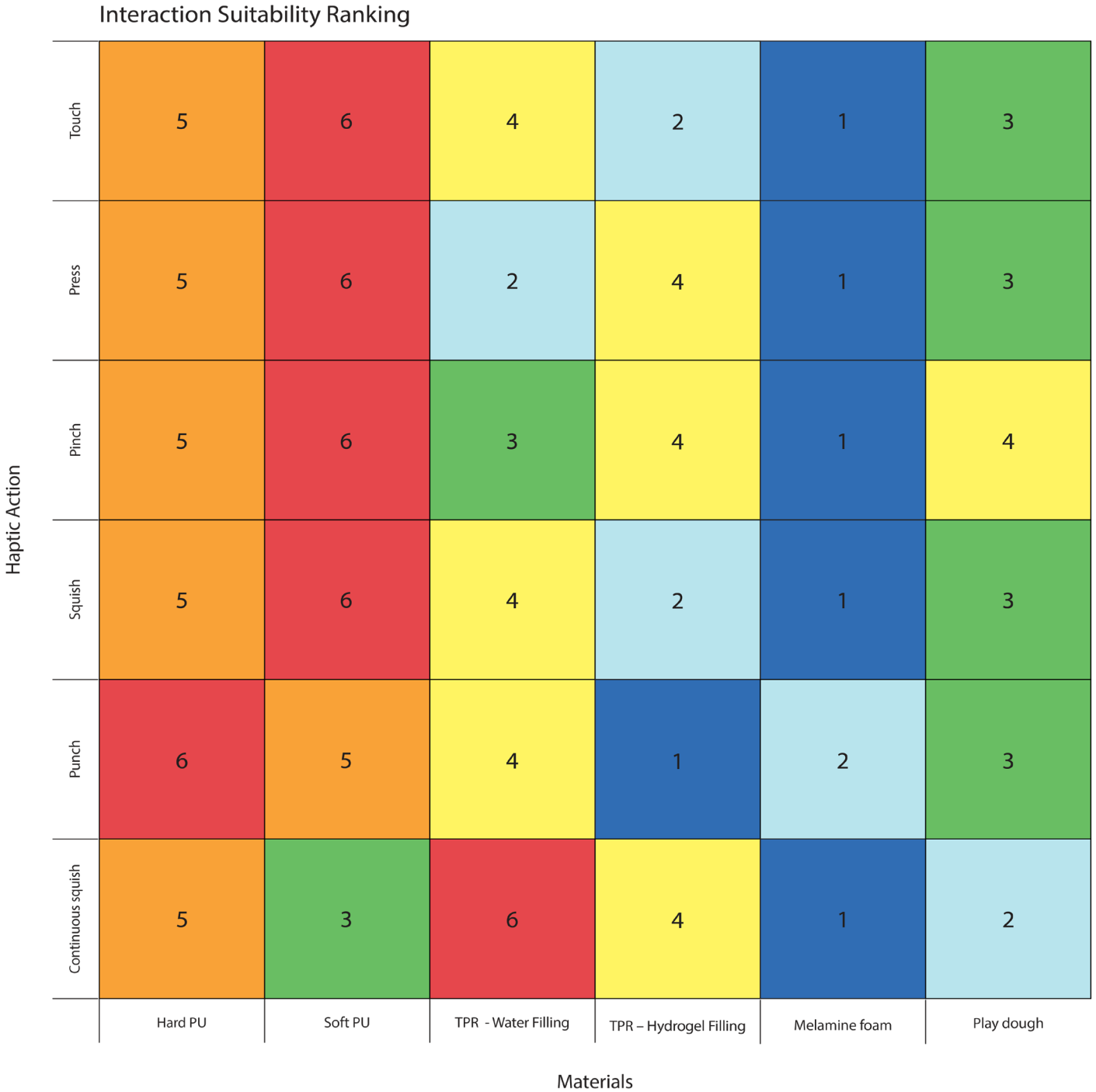


Figure 20. Rader chart that analyse the materials

make fast interaction physically demanding, while too little resistance can reduce tactile feedback. Similarly, very strong rebound or extremely fast recovery may make the interaction feel overly reactive, while very slow recovery can limit repeated use. Moderate responsiveness therefore provides a better balance between noticeable feedback and controllability.

The analysis also suggests that a large deformation range is not necessary for strong overall performance. The highest-performing materials tend to deform enough to communicate softness and pressure response without becoming excessively unstable or difficult to control. Based on these patterns, the ideal all-round squishy would combine high perceived softness and cushioning, low surface friction, moderate resistance and rebound, controlled recovery, moderate force sensitivity, and a contained deformation range. This indicates that the most effective tactile experience is created through a balance of material behaviours rather than maximising any single characteristic.



	Hard PU	Soft PU	TPR - Water Filling	TPR – Hydrogel Filling	Melamine foam	
Total	31	32	23	17	7	18

Figure 21. Heatmap that ranks the material performance

Most ideal Squishy
The target ranges were set according to the performance patterns observed across the six materials: Deformation Softness (5–6) because the best-performing materials were consistently soft and yielding; Compression Resistance (2–4) because too much resistance increased effort while too little reduced tactile feedback; Bounce/Elasticity (3–5) because moderate rebound provided responsiveness without becoming overly reactive; Cushioning (5–6) because stronger cushioning was associated with more comfortable impact absorption; Recovery Speed (2–4) because controlled recovery preserved feedback without limiting repeated interaction; Deformation Range (2–4) because moderate deformation gave clear tactile response while remaining controllable; Surface Friction (1–2) because lower friction reduced drag during repeated manipulation; and Speed Sensitivity (2–4) because moderate sensitivity allowed the material to respond to different interaction intensities without becoming unexpectedly firm.

Among the six materials tested, Soft PU is the strongest baseline material because it achieved the highest overall interaction-suitability score and already matches several of the desired characteristics, particularly high softness, strong cushioning and relatively low resistance. Its main limitation is that its rebound and recovery are comparatively weak, which makes it more suitable for slower, calming interactions than rapid repeated manipulation. Therefore, the most promising next step would be to retain the softness and cushioning qualities of Soft PU while slightly increasing its elastic rebound and recovery speed.

For further material development, I would therefore explore either a more responsive Soft PU formulation or a layered structure that combines a soft outer PU with a slightly more elastic internal layer. This could preserve the yielding and cushioned feeling that performed well in the current testing while improving responsiveness during repeated interaction. Rather than describing this as a "perfect" material, it would be more accurate to define it as an ideal target profile derived from the current comparative material exploration.

5.2 Colour Testing

I used a process of elimination to develop the final colour palette (see figure 22). I first grouped colours into five families—red, green, blue, yellow, and pink/purple—with nine variations in each group. From each family, I selected four preferred colours and brought them into the next stage, where all selected colours were mixed together to test how cohesively they worked as a complete palette.

I completed four rounds of elimination, removing the colours that felt most visually disconnected from the overall set. The aim was not to avoid strong accent colours, but to remove colours that dominated in a way that disrupted the balance of the palette. Through this process, I reduced the selection to eight final colours. These colours are intentionally not at maximum saturation, allowing them to remain joyful and expressive while still working cohesively across both the virtual workspace environment and the UX interface.

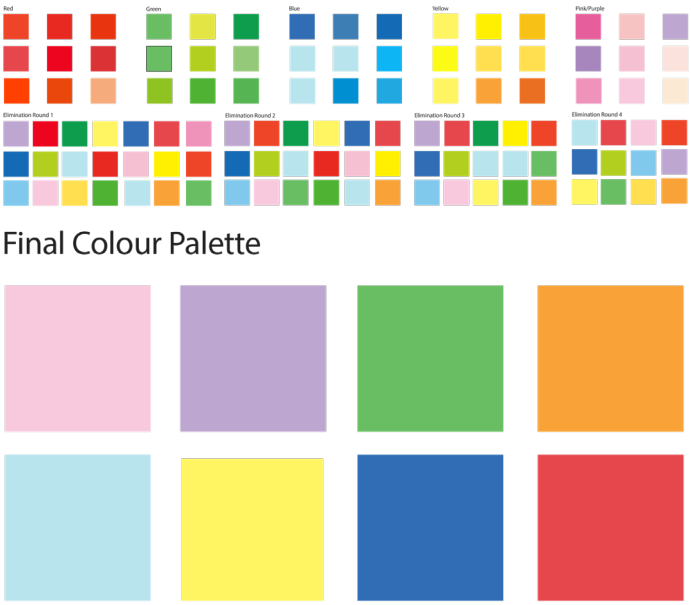


Figure 22. Final colour palette and the process of choosing the colour palette.

5.3 Everyday Sensory Exploration

Following the squishy comparison, I explored everyday actions and sensory experiences to identify possibilities beyond representing squishy objects on screen. The exploration considered how movement, deformation and sound could inform feedback for specific reading functions. Photographs and reflective notes documented these associations, which were then mapped to potential interface interactions (see Figure 23).

For button responses, squeezing, popping bubble wrap, operating switches and pressing keyboard keys offered contrasting feedback qualities. Soft deformation suggested a gentle response for frequently used controls, while sharper popping or clicking suggested more distinctive feedback for occasional actions. This comparison highlighted the importance of matching feedback intensity to the frequency and purpose of an interaction.

For search and AI assistance, I explored water-related references, including flowing water, needle ice and rain striking nearby surfaces. A small water-flow test produced less audible feedback than anticipated, whereas rain striking the roof and metal surfaces sounded particularly crisp and refreshing to me. This suggested that a digital interpretation could selectively combine visual movement with amplified or adapted sound, rather than reproduce the physical event literally.

For highlighting, copying and pasting, opening packaging, peeling tape and stretching a squishy

provided references for continuous movement and changing resistance. Pulling a packaging strip suggested a directional action similar to selecting a passage, while peeling tape offered an association with lifting and transferring selected content. Stretching and recovery suggested how a highlight or selection indicator might deform in response to dragging.

For reading-ruler interactions, I explored pencil and drawer movements as references for guiding an object along a controlled path. These suggested how a digital reading guide might move continuously through text, with visual or auditory feedback accompanying its movement. The exploration focused on giving the guide a tangible interaction quality while preserving its function of indicating reading position.

The exploration broadened the design direction from digital squishy objects towards interactions informed by everyday sensory qualities. It also suggested that satisfaction could arise from the relationship between movement, timing and sound, rather than appearance alone. These observations informed subsequent digital experiments; they represent my situated sensory impressions rather than evidence of improved reading or reduced stress.

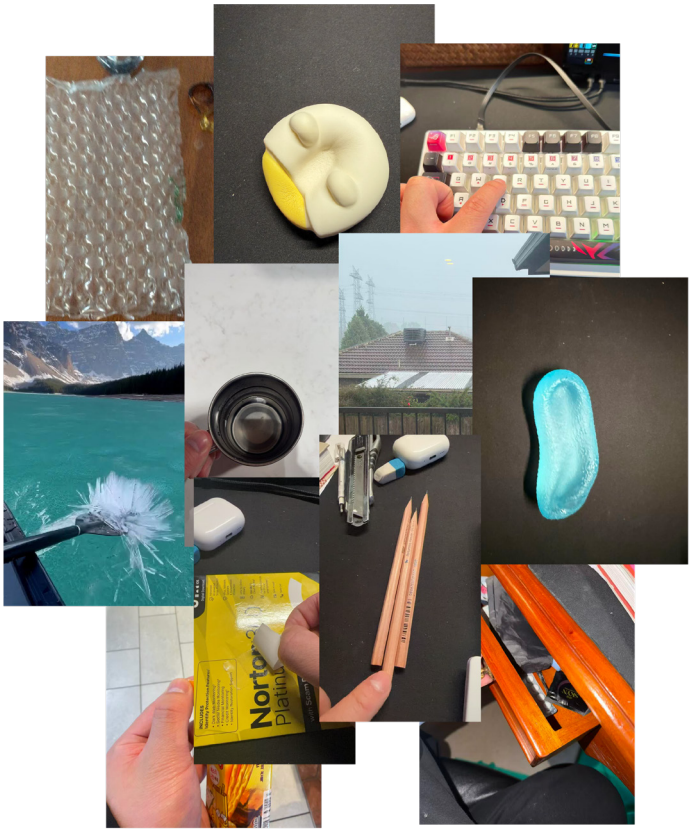


Figure 23. Collage of images showcase exploration.

5.4 Blender Modeling Testing

Building on the squishy material experiments and everyday sensory exploration, I used Blender to translate observed behaviours and photographic references into digital forms. Beginning with simple models, I explored deformation and recovery (see Figure 24), then applied different colours and textures to investigate how surface appearance could reinforce the intended material qualities (see Figure 25). At the end I modeled refined objects in the blender to use it later on for website building (see Figure 26)These trials progressed towards an animated interaction intended for integration into the website.

Alongside modelling, I examined further research to critically assess whether the emerging direction had a relevant empirical basis. Blanc et al. (2024) found that squeezing a vibrating stress ball reduced anxiety and, following stressful images, physiological arousal. Participants also rated vibration frequencies below their spontaneous squeezing frequency as more relaxing. This informed my exploration of slower deformation and recovery, while recognising that visual animation does not reproduce physical haptic feedback.

Andrade's (2010) study provided a complementary reference for exploring simple, repetitive interaction. Participants who shaded shapes while listening to a monotonous message recalled 29% more information than those who did not. This encouraged consideration of self-paced digital interactions with limited complexity, alongside scrutiny of whether their visual demands might compete with reading. The finding concerned an auditory task and therefore offered a rationale for exploration rather than evidence of equivalent benefits in this project.

The modelling and literature investigation together refined the direction towards repeatable interactions with considered timing, material appearance and feedback. The resulting animation brought these qualities into a form that could be explored within the reading interface.

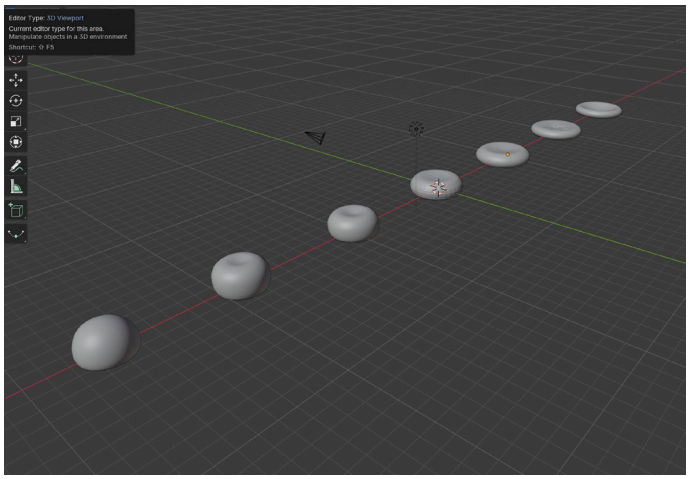


Figure 24. Modeling exploration

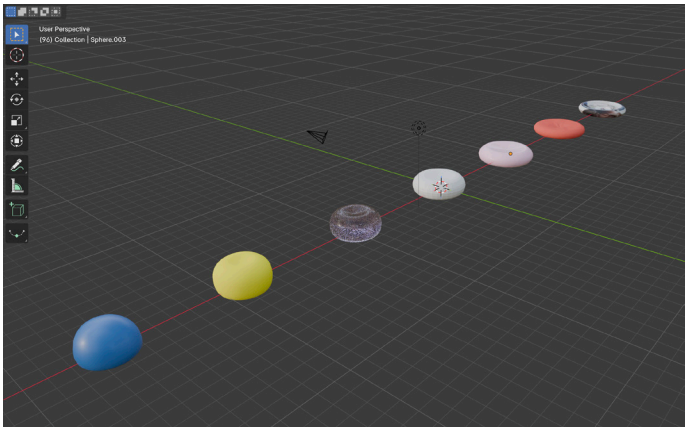


Figure 25. Materail rendering exploration.

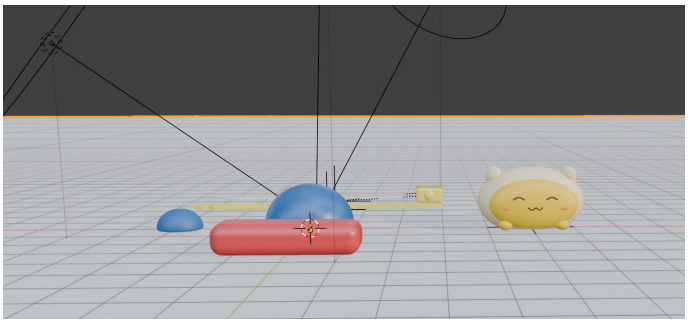
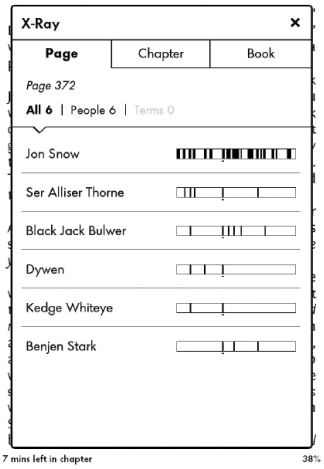
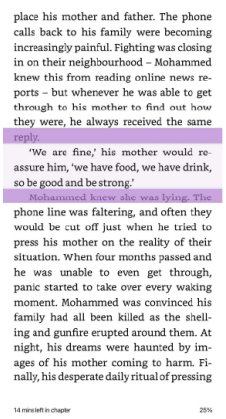


Figure 26. Refined modeling that will be used for on screen cue building.

APENDIX 1



Kindle has a feature called X-Ray. When the reader presses and holds a character, topic, event, place or term, X-Ray provides additional contextual information. It can also show where that character or term appears elsewhere in the book. This is the closest commercial precedent to your idea of selecting a key concept and exploring its relationships across a reading. Amazon states that X-Ray entries can be generated from book excerpts, public sources and machine learning.



place his mother and father. The phone calls back to his family were becoming increasingly painful. Fighting was closing in on their neighbourhood – Mohammed knew this from reading online news reports – but whenever he was able to get through to his mother to find out how they were, he always received the same reply.

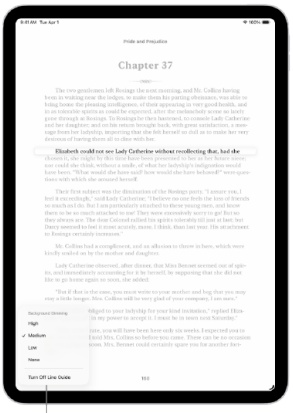
'We are fine,' his mother would reassure him, 'we have food, we have drink, so be good and be strong.'

Mohammed knew she was lying. The phone line was faltering, and often they would be cut off just when he tried to press his mother on the reality of their situation. When four months passed and he was unable to even get through, panic started to take over every waking moment. Mohammed was convinced his family had all been killed as the shelling and gunfire erupted around them. At night, his dreams were haunted by images of his mother coming to harm. Finally, his desperate daily ritual of pressing

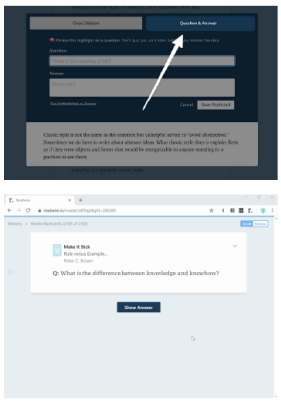
14 mins left in chapter 25%

Reading Ruler places a coloured visual guide over the current text to help the reader maintain their reading position. It is primarily an attention/ orientation aid, not automatic pacing. Amazon allows its colour, opacity, style and size to be customised.

This is a particularly useful precedent because it demonstrates the principle of reducing visual competition from surrounding text. Your design can build on this by adding controlled forward movement rather than requiring the reader to reposition the guide manually.



Apple has a feature named Line Guide. It makes one line more visually prominent while dimming the surrounding page. The reader can tap above or below the guide or drag it manually to another line. Therefore, like Kindle's Reading Ruler, this is guided attention rather than automatic pacing though. Also have very differenr phlopsy than Kindle where one try try to use colour to zone in a area you will foucs on and on the other side apple uses contrast to fade the other content to let you focus on particular one.



Readwise has a called Mastery, rather than simply Daily Review. Readwise Mastery turns saved highlights into active-recall interactions, particularly Question & Answer and Cloze Deletion. The reader must attempt to retrieve the information before revealing the answer; the cards are then resurfaced later through spaced repetition. This is a strong example for my retrieval cue.

APENDIX 2

Another noticeable feature is the frequent use of inflated typography, which is a major characteristic of Hypergoo and related visual styles. I think its main purpose is to make typography feel more lively, imperfect and object-like, allowing it to work more naturally with the 3D graphic elements. Script, serif and other decorative typefaces are also commonly used, including monospaced styles that reference early computer aesthetics and Y2K design. The examples marked with red lines show that when this expressive typography is missing, the design can lose a noticeable amount of energy.

However, this creates an important limitation for my project: these highly decorative typefaces can be difficult to read and are not suitable for large amounts of reading content. The examples marked with blue lines suggest a useful solution. I could reserve textured or 3D typography for more UX-heavy interactions, such as squishy elements or rewards, while using decorative type for section titles and maintaining a conventional sans-serif typeface for the main body text.

The use of noise and grain is another key visual effect I identified. It appears to serve three main purposes. First, it reinforces the nostalgic Y2K aesthetic. Second, adding a subtle grey texture over saturated colours can make them feel less intense and visually softer. Third, it introduces texture to the canvas, creating contrast with otherwise smooth and flat digital surfaces.

This is particularly useful for my project. The grain can create a slightly paper-like quality, making the screen feel less digitally flat. It also supports my intention to use colourful elements in a more controlled and comfortable way for longer reading sessions. Most importantly, using grain consistently across the interface could help connect different visual elements and strengthen the overall design language.

One minor but important characteristic I noticed is how this visual style reduces the visible presence of the grid. This does not mean there is no grid or structure, but the grid is made less obvious so that images and typography can feel more flexible. This is often achieved by tilting images or letters, varying their scale, and cropping objects out of their original image backgrounds so they can be positioned more freely.

When some elements still need to stay within a grid for clarity, irregular shapes are often used instead of standard rectangles. For example, folder-like or asymmetrical containers can still organise content while making the layout feel less rigid.

This is useful for my own design because the reading section will inevitably need a grid for readability. These techniques could help me keep that structure while making the interface feel more natural, flexible and visually engaging.

Texture is one of the strongest characteristics of the design style I analysed. Whether metallic, foamy, plastic-like or rubbery, giving 3D objects a clear material quality helps make the design feel more lively and tactile. I think this is partly because the textured 3D objects create a strong contrast with the flatter typography and graphic elements around them. Their material detail also gives them more visual emphasis and makes them feel more physical rather than purely digital. The examples marked with the red line work on their own, but compared with the more textured references, they feel noticeably flatter and less distinctive.

For my own design, this reinforces that the squishy objects I develop should each have their own recognisable material texture rather than only differing in shape or movement. It also expands my thinking beyond squishy objects alone: other satisfying real-world materials or interactions could potentially be translated into digital form, creating a broader and more playful sensory experience.



Another recurring pattern is the use of cut-out objects instead of keeping images inside rectangular photo frames. Because the object itself has more complex and irregular edges, it creates a more lively and less restricted feeling. When placed within a grid, it also avoids the repetitive sense of every image being "kept in a box." The example marked with the red line feels noticeably more rigid and less free compared with the cut-out references.

This is useful for my own design because current UI often relies heavily on simple rectangular or circular containers. Introducing more irregular shapes, or allowing the shape of the object itself to define the visual boundary, could make the experience feel more playful and less purely digital. It may also help create a stronger bridge between digital and physical experience, since objects in the real world are not naturally contained within photo frames.

A small but critical finding is that, although this style can be categorised as relatively maximalist, it actually creates a careful balance between maximalism and minimalism. Even when there are many visual elements, such as expressive typography and images, the background is often kept very simple with plain colours rather than filling every available space. This shows that the balance between positive and negative space is still highly important.

This is especially relevant to my project because the reading interface will already contain a large amount of text. I therefore need to control the number of active visual elements on screen. By active, I mean elements that remain visible regardless of whether the user interacts with them. More passive functions, such as copy, highlight or other tools, can remain hidden until activated. This could help maintain a visually engaging style without making the reading experience feel overcrowded or distracting.

Another pattern I noticed is the use of relatively lower-saturation colours. By this, I do not mean grey or muted colours, but choosing a slightly less saturated version of an otherwise bright colour. The palette can still feel warm and visually strong. For example, instead of a pure bright red, an orange-red may be used, while a highly RGB-like green may shift towards a more yellow-green tone. This may partly reference a vintage aesthetic, but it also makes the colours feel less visually aggressive.

This reinforces why the style suits my project. Since the interface is designed for longer periods of reading, using the most saturated version of every colour could become tiring. The example marked with the red line is a useful negative reference for my intended direction: it successfully creates strong visual hierarchy, but its highly saturated colour would be less suitable for a long-term reading experience.

I also noticed that the palette generally leans towards warmer undertones. Even cooler colours such as green and blue are often shifted slightly towards yellow. This is useful when developing my own colour palette, as sharing a similar undertone could help different colours feel more cohesive while still allowing the interface to remain colourful and expressive.

One small pattern I noticed is the use of mixed typography styles within the same sentence or composition, such as combining a serif typeface with a curly script. I do not necessarily want to copy this exact technique, but it suggests two useful directions for my project.

First, because my design will likely use multiple typefaces, I should develop a cohesive type family or pairing system where different styles can work together without feeling disconnected. Second, more decorative typefaces could be reserved for interactive moments such as hover effects, rewards or transitions. This could allow typography itself to become part of the experience without affecting the readability of the main body text.

Key Finding 1 — Reduced Grid Visibility
Structure remains, but tilted elements, scale variation and irregular shapes make the layout feel less rigid.

Key Finding 2 — Cut-out Imagery
Removing rectangular image frames makes objects feel freer, more lively and less confined.

Key Finding 3 — Mixed Typography
Different type styles are combined to create more personality and visual energy.

Key Finding 4 — Controlled Colour Saturation
Colours stay warm and expressive, but avoid the most intense saturated versions for greater visual comfort.

Key Finding 5 — Balanced Maximalism
Rich visual elements are balanced with simple backgrounds and negative space.

Key Finding 6 — Material Texture
Metallic, foamy, rubbery and plastic-like textures make 3D elements feel more tactile and realistic.

Key Finding 7 — Inflated / Decorative Typography
Expressive typography adds energy, but should be used selectively rather than for long-form reading.

Key Finding 8 — Noise and Grain
Grain adds Y2K character, softens strong colours and gives the digital surface a more tactile quality.